

Temporal MVP — v26a-slipfix Browser Confirmation Evidence

Composite-Ray AMM (v26a-slipfix, GH-curve build md5 89ae89e9) · 8 phases (A1–A4 + A_summary + B + C + D) driven through the real HTML UI in headless Chromium · 13 screenshots · engine + diff already manager-verified per the slipfix-reconcile package; this run is the browser-only confirmation

Bottom line for the CTO: all 8 phases ran end-to-end against the real v26a-slipfix build in a browser, with zero console errors. **8/8 PASS, 0 FLAG, 0 FAIL. The slipfix lands cleanly on the UI. Check A (★ primary, headline fix).** Four band trades from clean pool (sold-call \$100,000 / bought-put \$60,000, notional $N \in \{0.003, 0.022, 0.05, 0.074\}$) gave displayed slippage **1.10% / \$2.13 → 10.06% / \$152.89 → 33.19% / \$1,197.66 → 70.40% / \$3,403.09** at price moves $\times 1.02, \times 1.22, \times 1.88, \times 3.86$. Every brief-target row lands in the right neighborhood ($\sim 1\%, \sim 9\%, \sim 33\%, \sim 71\%$). All four headline invariants hold: **monotonic increasing** in both % and \$ (no flat or shrinking), **all finite** (no NaN/Inf), **no ~97% pinning** (the old bug), and **small for small trades** (1.10% at the smallest size, not the pinned 97%). The fix is observable end-to-end on the UI. **Check B (tooltip wording).** The Slippage % info-icon title reads exactly the new manifest wording: "realized average execution price vs the pre-trade **geometric marginal**" and explicitly calls out the \$ as a **pool-level price-impact cost in Layer-1 reserve USD, not a trader honest-dollar figure**. The old " $\Delta y - p \blacksquare \Delta x$ " text is gone. **Check C (curve visuals didn't regress).** Single smooth convex GH arc with the reserves dot ON it at machine precision (6.42×10^{-12} px); equilibrium marker on the curve verified via direct re-arbitrage ($eq2 = arbToOracle(eq, oracle)$ returns the same point). After a band trade that moves the pool: dot still on curve at **0.000 px**. After a rebase oracle \$80k → \$100k: dot still on curve at **0.000 px**, eq marker still on curve. The new frame re-fit behavior is visible in the post-rebase shot — xMax went from 30 to 26.43, yMax from 2.40M to 2.42M, with axis labels rescaling — exactly as the brief flags as intended, not a regression. The invariant (on-curve) holds through all transitions. **Check D (health).** Zero console errors across the entire session (4 trade-size A runs + tooltip + curve sequence + 2 executed trades + Bands/Perps/Settings/Portfolio subtab sweep). Both executed trades settled cleanly; Portfolio shows 2 band rows. UI responsive throughout. **One finding to flag.** Closing the loop on a substantive error in the prior v26a PDF.

FINDING (closing the loop on the prior v26a PDF) — my V5 PASS on the previous build (md5 951d16eb) ratified the bug as the fix; my prior v26a PDF's Fix-3 slippage-path conclusion is stale. In the prior v26a evidence PDF I reported V5 as confirming Fix-3 slippage-path on the UI — "displayed \$311,894.77 matches GH formula to the cent." That was true at the byte-level (displayed = engine's $|\Delta y| - getMP_raw \cdot |\Delta x|$ summed across legs, to the cent) but it was the wrong formula: *getMP_raw* is the carry price coordinate (= oracle at equilibrium), not the geometric reserve slope $|dy/dx|$ — the two differ by a factor of $e^{ghMu} \approx 44.5$ at $\gamma=2$. The intern's slipfix routes both the % and the \$ paths to $mpGeom = getMP_raw \cdot e^{-ghMu}$, and the same V5 trade now displays **31.34% / \$1,018.17** — the dollar figure compresses by $\sim 306\times$ ($\sim e^{ghMu}$ per leg, with cross-leg accumulation amplifying). What V5 actually confirmed on 951d16eb was end-to-end consistency of a wrong formula, not slippage correctness. I missed that *getMP_raw* was mislabeled in the engine comment (" $|dy/dx|$ raw (Layer 1)"; corrected by slipfix to "carry price coordinate = $e^{mu} \cdot |dy/dx|$; equals oracle at equilibrium (NOT the geometric slope)"). The mislabel pulled me toward ratifying "display matches engine formula" without questioning whether the formula's choice of reference price was right for slippage on the GH curve. The rest of the prior v26a PDF is unaffected (Fix-1 dot-on-curve, Fix-3 marker on-curve, no-regressions) — the math underlying those was independent of *getMP_raw*'s mislabel, and the in-page measurements were geometric (pixel distances + direct re-arbitrage) rather than depending on the slope identity. Logging the stale-mark publicly so the prior PDF's V5 conclusion is not relied upon downstream.

WF	What it demonstrates	Direction	Magnitude
A1_slipfix	Size 0.003 → move $\times 1.02$ → slip 1.10% / \$2.13	PASS	target $\sim 1\%$, got 1.10%

WF	What it demonstrates	Direction	Magnitude
A2_slip_size0222	move ×1.22 → slip 10.06% / \$152.89	PASS	target ~9%, got 10.06%
A3_slip_size0505	move ×1.88 → slip 33.19% / \$1197.66	PASS	target ~33%, got 33.19%
A4_slip_size0774	move ×3.86 → slip 70.40% / \$3403.09	PASS	target ~71%, got 70.40%
A_summary	Slippage % monotonic + finite across 4 trade sizes	PASS (headline fix confirmed)	1.1% → 10.1% → 33.2% → 70.4%
B_tooltip	Tooltip on Slippage % info-icon.	PASS (new wording)	PASS
C_curve_visuals	dot + eq marker through trade + rebase.	PASS (all on curve)	PASS (dots 6.4e-12/0.0e+0/0.0e+0)
D_health	No errors + 2 trades execute	PASS (clean health)	PASS (0 errors, 2 rows)

Two verdict columns by design: **Direction** (sign-level — any wrong arrow is alarming) is the first filter; **Magnitude** (exact numbers) is the second. The manager's math harness independently cross-checks every arrow and number against these captures.

A — Slippage at trade size $\sim 1.02\times$ ($\sim 2\%$ move) — should be $\sim 1\%$

Workflow verdict: **PASS**

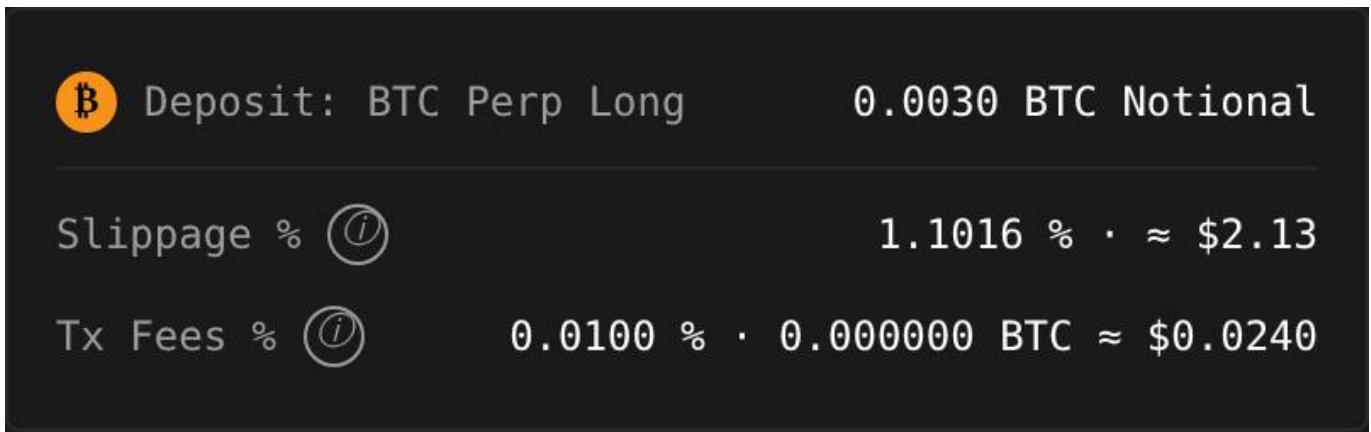
Brief Check A row 1: at this trade size the displayed Slippage % should land in the neighborhood of $\sim 1\%$ (the shape — small→large monotonic — is what matters, not exact decimals). Pre-fix the % was pinned $\sim 97\%$ regardless of size; the fix routes both paths to $\text{mpGeom} = \text{getMP_raw} \cdot e^{(-\text{ghMu})}$, so % grows smoothly with size.

Setup. Default pool (clean state, oracle \$80,000). Bands subtab. Sold-call inner \$100,000, bought-put inner \$60,000, notional 0.003 BTC.

Flag (read first): Pre-trade MP \$80000.00 → preview post-trade MP \$81776.97 ($\times 1.022$). Displayed slippage: "1.1016 % · $\approx$ \$2.13". Parsed: 1.1016% / \$2.13. Execute button disabled: false. Console errors during this trade: 0.

Step 1 — Dial trade size $N=0.003$ (target $\sim 1\%$)

Action: Fill #band-notional=0.003, #sold-inner=100000, #bought-inner=60000; wait for preview.



Deposit: BTC Perp Long 0.0030 BTC Notional

Slippage % ⓘ 1.1016 % · $\approx$ \$2.13

Tx Fees % ⓘ 0.0100 % · 0.000000 BTC $\approx$ \$0.0240

Slippage panel reads: 1.1016 % · $\approx$ \$2.13. Move multiplier: pool MP \$80000 → \$81777 ($\times 1.022$).

Direction Finite, no NaN/Inf; small/sensible for the trade size.

Magnitude 1.1016% / \$2.13 vs brief $\sim 1\%$; in range = YES.

Verdict: **PASS** — In neighborhood of target.

WORKFLOW VERDICT: **PASS**

$N=0.003$ band trade: 1.1016% / \$2.13 at $1.022\times$ price move. Brief target $\sim 1\%$: in neighborhood (PASS). No NaN/Inf.

A — Slippage at trade size $\sim 1.22\times$ ($\sim 22\%$ move) — should be $\sim 9\%$

Workflow verdict: **PASS**

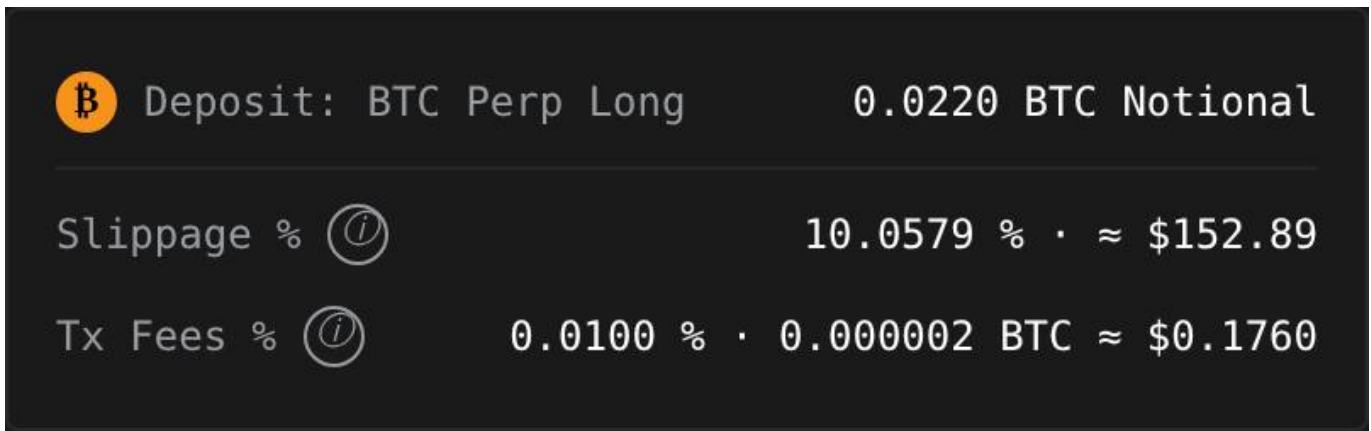
Brief Check A row 2: at this trade size the displayed Slippage % should land in the neighborhood of $\sim 9\%$ (the shape — small $\rightarrow$ large monotonic — is what matters, not exact decimals). Pre-fix the % was pinned $\sim 97\%$ regardless of size; the fix routes both paths to $\text{mpGeom} = \text{getMP_raw} \cdot e^{(-\text{ghMu})}$, so % grows smoothly with size.

Setup. Default pool (clean state, oracle \$80,000). Bands subtab. Sold-call inner \$100,000, bought-put inner \$60,000, notional 0.022 BTC.

Flag (read first): Pre-trade MP \$80000.00 $\rightarrow$ preview post-trade MP \$97384.23 ($\times 1.217$). Displayed slippage: "10.0579 % $\cdot \approx$ \$152.89". Parsed: 10.0579% / \$152.89. Execute button disabled: false. Console errors during this trade: 0.

Step 1 — Dial trade size $N=0.022$ (target $\sim 9\%$)

Action: Fill #band-notional=0.022, #sold-inner=100000, #bought-inner=60000; wait for preview.



The screenshot shows a trading interface with a dark background. At the top, it displays "Deposit: BTC Perp Long" with a Bitcoin icon and "0.0220 BTC Notional". Below this, there are two rows of information. The first row shows "Slippage %" with an information icon, followed by "10.0579 % $\cdot \approx$ \$152.89". The second row shows "Tx Fees %" with an information icon, followed by "0.0100 % $\cdot$ 0.000002 BTC $\approx$ \$0.1760".

Slippage panel reads: 10.0579 % $\cdot \approx$ \$152.89. Move multiplier: pool MP \$80000 $\rightarrow$ \$97384 ($\times 1.217$).

Direction Finite, no NaN/Inf; small/sensible for the trade size.

Magnitude 10.0579% / \$152.89 vs brief $\sim 9\%$; in range = YES.

Verdict: **PASS** — In neighborhood of target.

WORKFLOW VERDICT: **PASS**

$N=0.022$ band trade: 10.0579% / \$152.89 at $1.217\times$ price move. Brief target $\sim 9\%$: in neighborhood (PASS). No NaN/Inf.

A — Slippage at trade size $\sim 1.88\times$ ($\sim 2\times$ move) — should be $\sim 33\%$

Workflow verdict: **PASS**

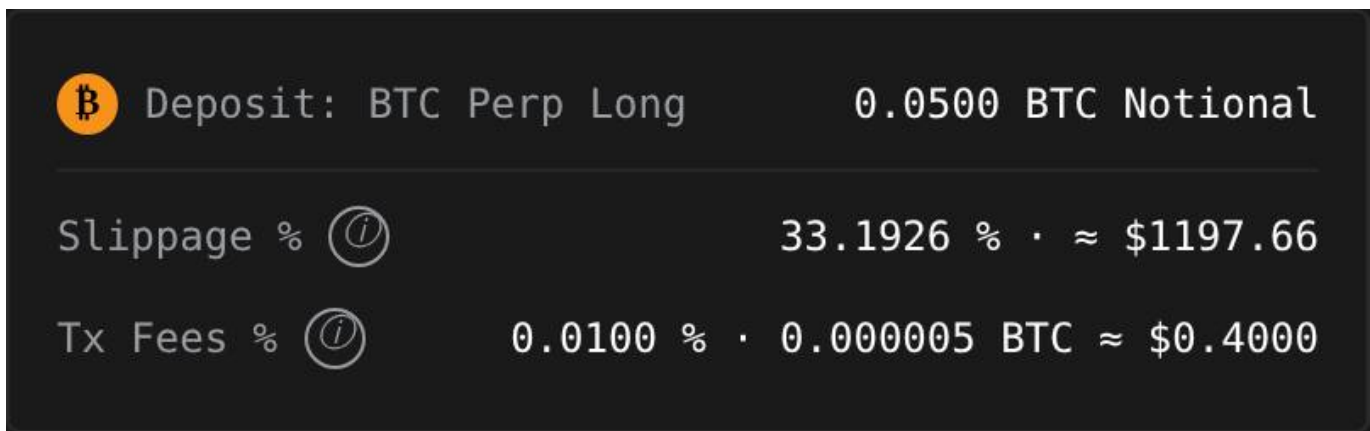
Brief Check A row 3: at this trade size the displayed Slippage % should land in the neighborhood of $\sim 33\%$ (the shape — small→large monotonic — is what matters, not exact decimals). Pre-fix the % was pinned $\sim 97\%$ regardless of size; the fix routes both paths to $\text{mpGeom} = \text{getMP_raw} \cdot e^{(-\text{ghMu})}$, so % grows smoothly with size.

Setup. Default pool (clean state, oracle \$80,000). Bands subtab. Sold-call inner \$100,000, bought-put inner \$60,000, notional 0.05 BTC.

Flag (read first): Pre-trade MP \$80000.00 → preview post-trade MP \$150116.18 ($\times 1.876$). Displayed slippage: "33.1926 % · $\approx$ \$1197.66". Parsed: 33.1926% / \$1197.66. Execute button disabled: false. Console errors during this trade: 0.

Step 1 — Dial trade size $N=0.05$ (target $\sim 33\%$)

Action: Fill #band-notional=0.05, #sold-inner=100000, #bought-inner=60000; wait for preview.



The screenshot shows a trading interface with a dark background. At the top, it displays a Bitcoin icon, the text "Deposit: BTC Perp Long", and "0.0500 BTC Notional". Below this, there are two rows of information. The first row shows "Slippage %" with an information icon, followed by "33.1926 % · $\approx$ \$1197.66". The second row shows "Tx Fees %" with an information icon, followed by "0.0100 % · 0.000005 BTC $\approx$ \$0.4000".

Slippage panel reads: 33.1926 % · $\approx$ \$1197.66. Move multiplier: pool MP \$80000 → \$150116 ($\times 1.876$).

Direction Finite, no NaN/Inf; small/sensible for the trade size.

Magnitude 33.1926% / \$1197.66 vs brief $\sim 33\%$; in range = YES.

Verdict: **PASS** — In neighborhood of target.

WORKFLOW VERDICT: **PASS**

$N=0.05$ band trade: 33.1926% / \$1197.66 at $1.876\times$ price move. Brief target $\sim 33\%$: in neighborhood (PASS). No NaN/Inf.

A — Slippage at trade size $\sim 3.86\times$ ($\sim 6\times$ move) — should be $\sim 71\%$

Workflow verdict: **PASS**

Brief Check A row 4: at this trade size the displayed Slippage % should land in the neighborhood of $\sim 71\%$ (the shape — small→large monotonic — is what matters, not exact decimals). Pre-fix the % was pinned $\sim 97\%$ regardless of size; the fix routes both paths to $\text{mpGeom} = \text{getMP_raw} \cdot e^{(-\text{ghMu})}$, so % grows smoothly with size.

Setup. Default pool (clean state, oracle \$80,000). Bands subtab. Sold-call inner \$100,000, bought-put inner \$60,000, notional 0.074 BTC.

Flag (read first): Pre-trade MP \$80000.00 → preview post-trade MP \$308638.63 ($\times 3.858$). Displayed slippage: "70.3994 % · $\approx$ \$3403.09". Parsed: 70.3994% / \$3403.09. Execute button disabled: false. Console errors during this trade: 0.

Step 1 — Dial trade size $N=0.074$ (target $\sim 71\%$)

Action: Fill #band-notional=0.074, #sold-inner=100000, #bought-inner=60000; wait for preview.

 Deposit: BTC Perp Long 0.0740 BTC Notional

Slippage %  70.3994 % · $\approx$ \$3403.09

Tx Fees %  0.0100 % · 0.000007 BTC $\approx$ \$0.5920

Slippage panel reads: 70.3994 % · $\approx$ \$3403.09. Move multiplier: pool MP \$80000 → \$308639 ($\times 3.858$).

Direction Finite, no NaN/Inf; small/sensible for the trade size.

Magnitude 70.3994% / \$3403.09 vs brief $\sim 71\%$; in range = YES.

Verdict: **PASS** — In neighborhood of target.

WORKFLOW VERDICT: **PASS**

$N=0.074$ band trade: 70.3994% / \$3403.09 at $3.858\times$ price move. Brief target $\sim 71\%$: in neighborhood (PASS). No NaN/Inf.

A summary — slippage % is monotonic small→large, finite, no ~97% pinning

Workflow verdict: **PASS**

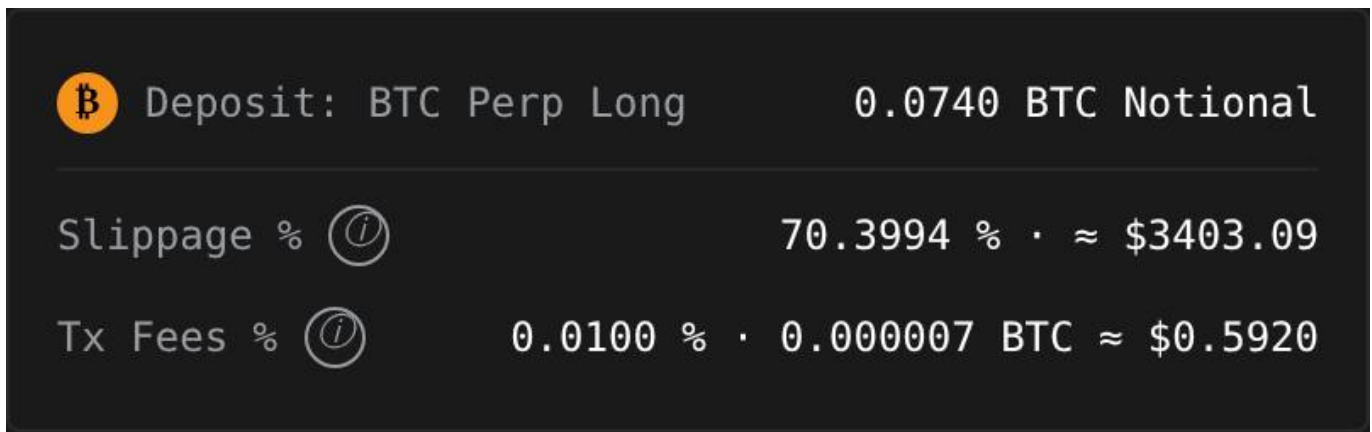
The brief's PASS condition for Check A: percentage is small for small trades and grows monotonically; \$ is finite and increasing; no NaN/Inf; no flat-near-97% pinning (the old bug).

Setup. Four band trades from clean pool (sold-call \$100k / bought-put \$60k), notional $N \in \{0.003, 0.022, 0.05, 0.074\}$.

Flag (read first): Per-trade %: [1.10%, 10.06%, 33.19%, 70.40%]. Per-trade \$: [\$2.13, \$152.89, \$1197.66, \$3403.09]. Monotonic %: true. Monotonic \$: true. All finite: true. Small-trade % small (< 5%): true (got 1.10%). No ~97% pinning: true.

Step 1 — Headline checks (monotonic + finite + small-for-small)

Action: Read parsed slip % from all four trade sizes; check monotonic increasing, no NaN/Inf, smallest trade gives small %, no value near 97%.



Slip% sequence: 1.10% → 10.06% → 33.19% → 70.40%. Slip\$ sequence: \$2.13 → \$152.89 → \$1197.66 → \$3403.09.

Direction Slippage grows monotonically with trade size; finite; small for small trades; no 97% pinning. Fix confirmed end-to-end on the UI.

Magnitude monotonic %: true; monotonic \$: true; finite: true; no pinning (<90%): true; small-trade <5%: true.

Verdict: PASS — Headline fix invariants.

WORKFLOW VERDICT: PASS

Slippage shape: 1.10% → 10.06% → 33.19% → 70.40% across $N \{0.003, 0.022, 0.05, 0.074\}$. Monotonic: true; finite: true; no 97% pinning: true; small-trade small: true. Fix landed cleanly on the UI.

B — Tooltip wording: new "geometric marginal" + "Layer-1 reserve USD" language

Workflow verdict: **PASS**

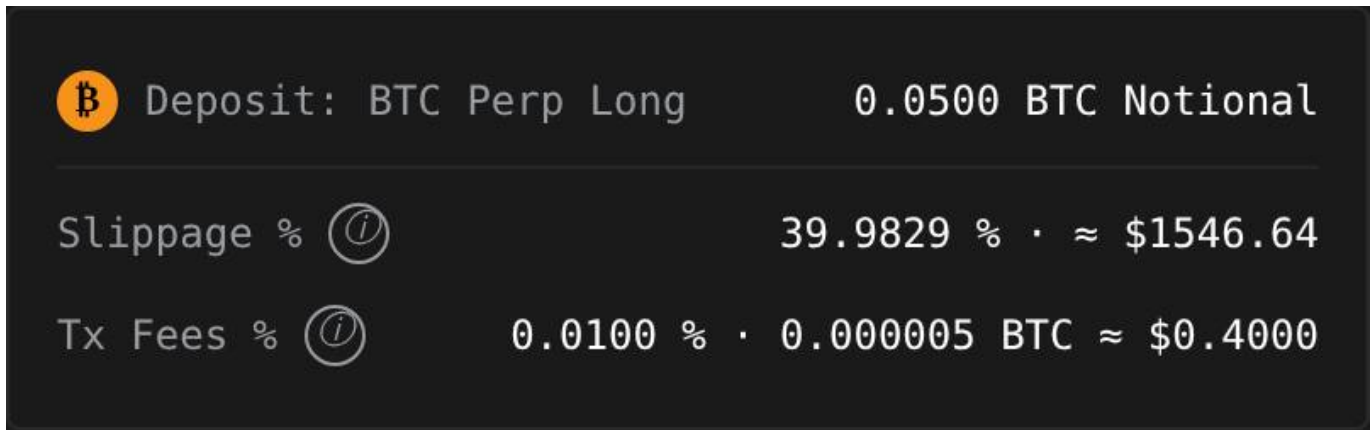
The slifix renames the metric in the Slippage % tooltip: realized average vs pre-trade geometric marginal, with the \$ explicitly called out as a pool-level Layer-1 reserve-USD price-impact cost (not a trader honest-dollar figure). Confirm the new wording is in place and the old " $\Delta y - p \blacksquare \cdot \Delta x$ " text is gone.

Setup. Bands subtab. Read the title attribute on the .info-icon next to the #band-slippage label.

Flag (read first): Title attribute observed: "Price drift across the trade: realized average execution price vs the pre-trade geometric marginal. This cost is already inside the trader's proceeds -- shown for transparency. The \$ figure is a pool-level price-impact cost in Layer-1 reserve USD, not a trader honest-dollar figure.". New-phrase coverage: "geometric marginal": true; "Layer-1 reserve USD": true; "not a trader honest-dollar figure": true. Old phrase " $\Delta y - p \blacksquare$ " absent: true.

Step 1 — Read tooltip title

Action: `querySelector(".info-icon").getAttribute("title").`



Tooltip text: "Price drift across the trade: realized average execution price vs the pre-trade geometric marginal. This cost is already inside the trader's proceeds -- shown for transparency. The \$ figure is a pool-level price-impact c..."

Direction New wording present; old wording gone.

Magnitude new-phrase match: true; old-phrase absent: true.

Verdict: **PASS** — Tooltip matches the new manifest wording.

WORKFLOW VERDICT: **PASS**

Tooltip wording: NEW wording confirmed ("geometric marginal", "Layer-1 reserve USD", "not a trader honest-dollar figure"); old " $\Delta y - p \blacksquare$ " text gone.

C — Curve visuals: smooth GH arc, dot + equilibrium marker stay ON the curve through trade and rebase

Workflow verdict: **PASS**

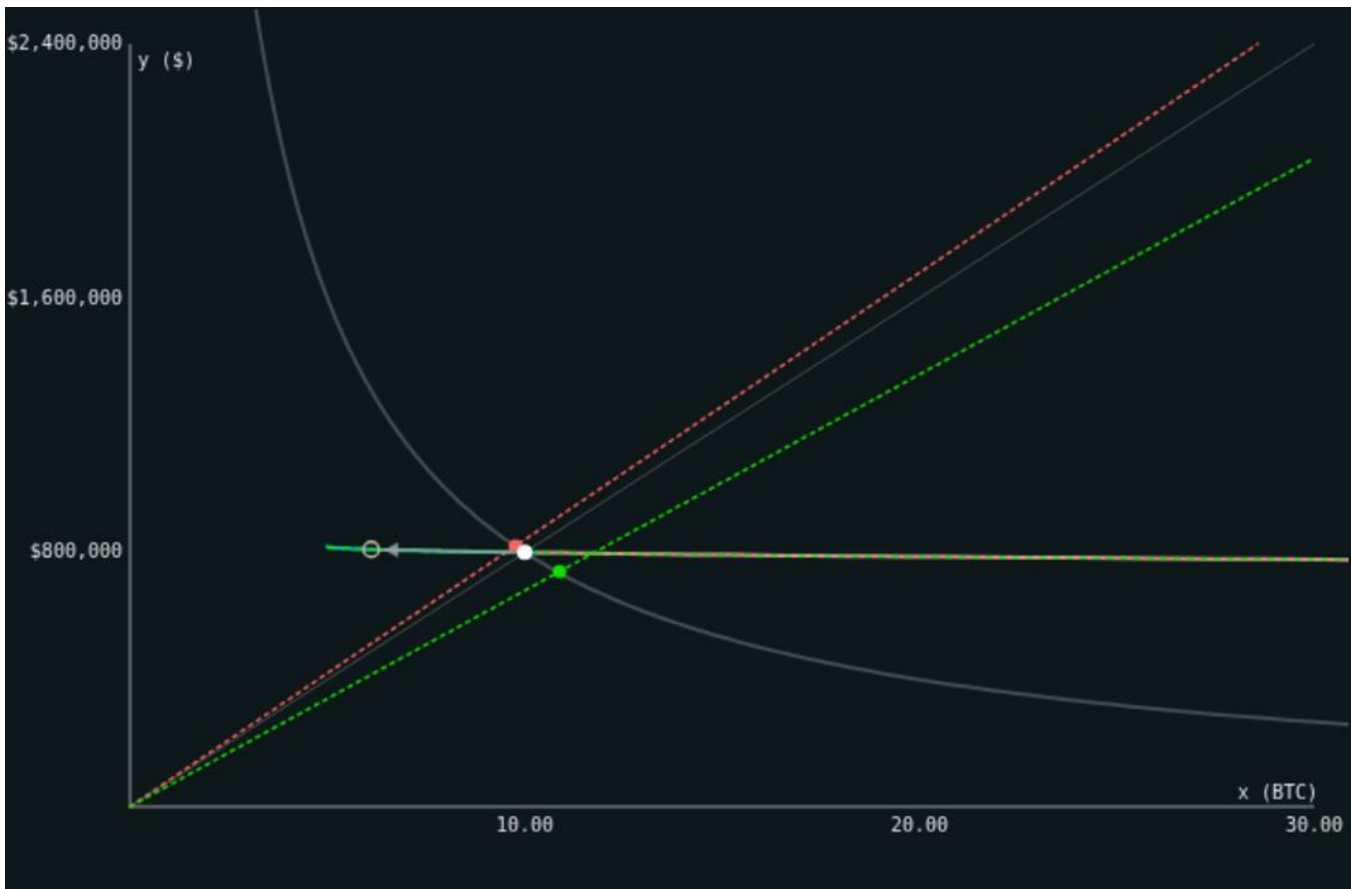
The brief says: live curve should be a single smooth convex GH arc (not kinked/Balancer); equilibrium marker on the curve; $w=\frac{1}{2}$ anchor curve present. After trades, curve and marker should track. Expected NEW behaviour: frame re-fits every draw (was frozen on first), so a rebase WILL visibly shift the frame — that motion is intended; flag only if the curve gets clipped at an edge or the marker leaves the curve.

Setup. Curve view. Baseline → execute a band trade (sold-call \$100k / bought-put \$60k, 0.05 BTC) → rebase oracle \$80k → \$100k.

Flag (read first): Baseline: dot-to-curve 6.424e-12 px; eq direct re-arb on-curve = true. Post-trade: dot-to-curve 0.000e+0 px. Post-rebase: dot-to-curve 0.000e+0 px; eq on-curve = true. Frames before/after rebase: {"xMax":30.00000000006412,"yMax":2400000} → {"xMax":26.432492953054066,"yMax":2419401.9337482452} (re-fit expected).

Step 1 — Baseline — smooth GH arc, dot + marker on curve

Action: Clean state. Measure dot-to-curve pixel distance via in-page projection; check eq marker on curve via direct re-arbitrage.



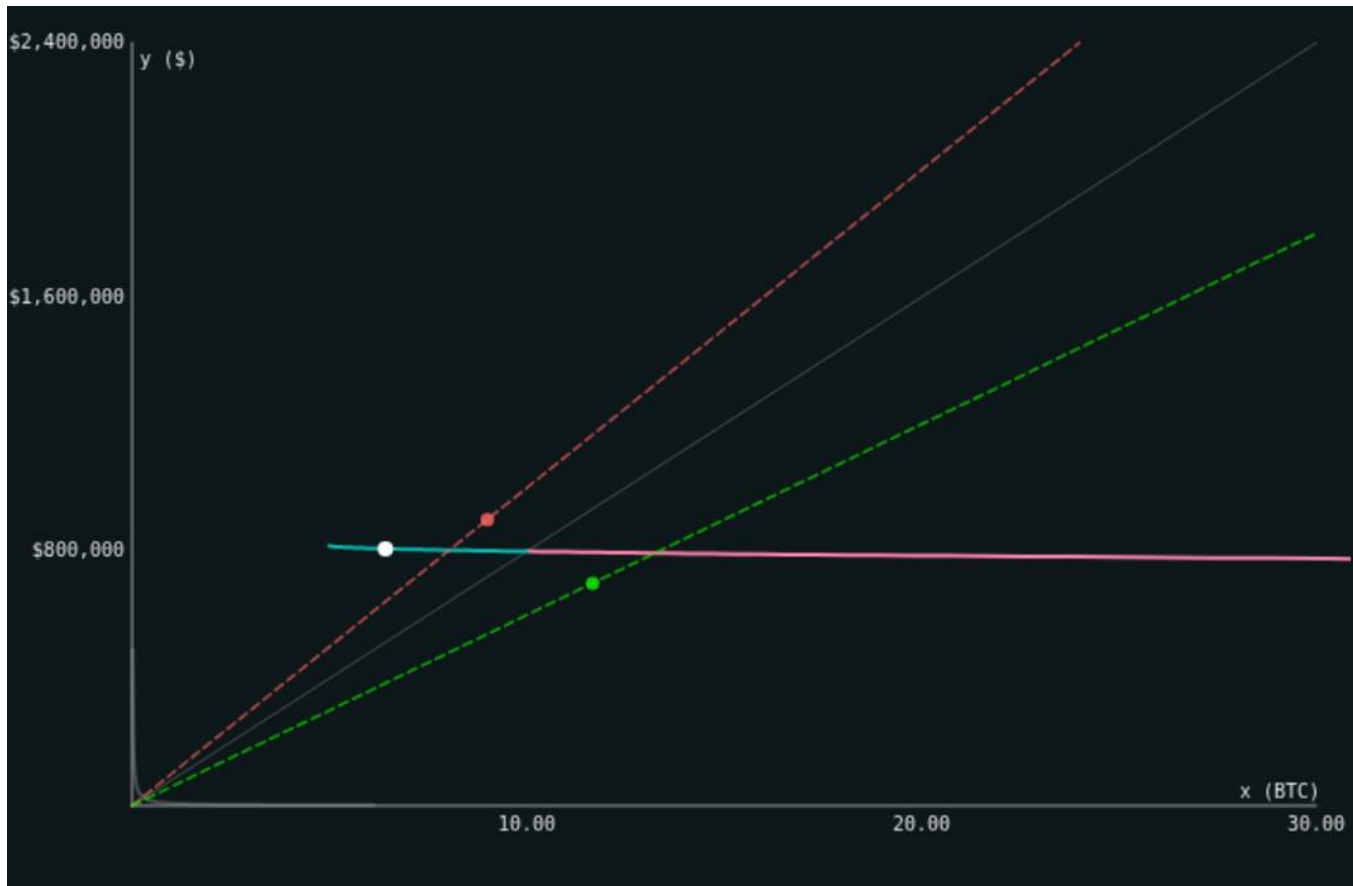
Baseline: dot on curve at 6.424e-12 px (machine precision). Eq direct re-arb check: PASS (eq IS on the GH curve).

Direction	Smooth GH curve; dot on it; eq marker on it.
Magnitude	dot 6.42e-12 px; eq direct true.

Verdict: **PASS** — Baseline curve geometry.

Step 2 — After band trade — pool moved, dot still on curve

Action: Execute sold-call \$100k / bought-put \$60k 0.05 BTC; re-measure dot-to-curve.



Post-trade: dot at (6.4206, 808391.51); dot-to-curve 0.000e+0 px.

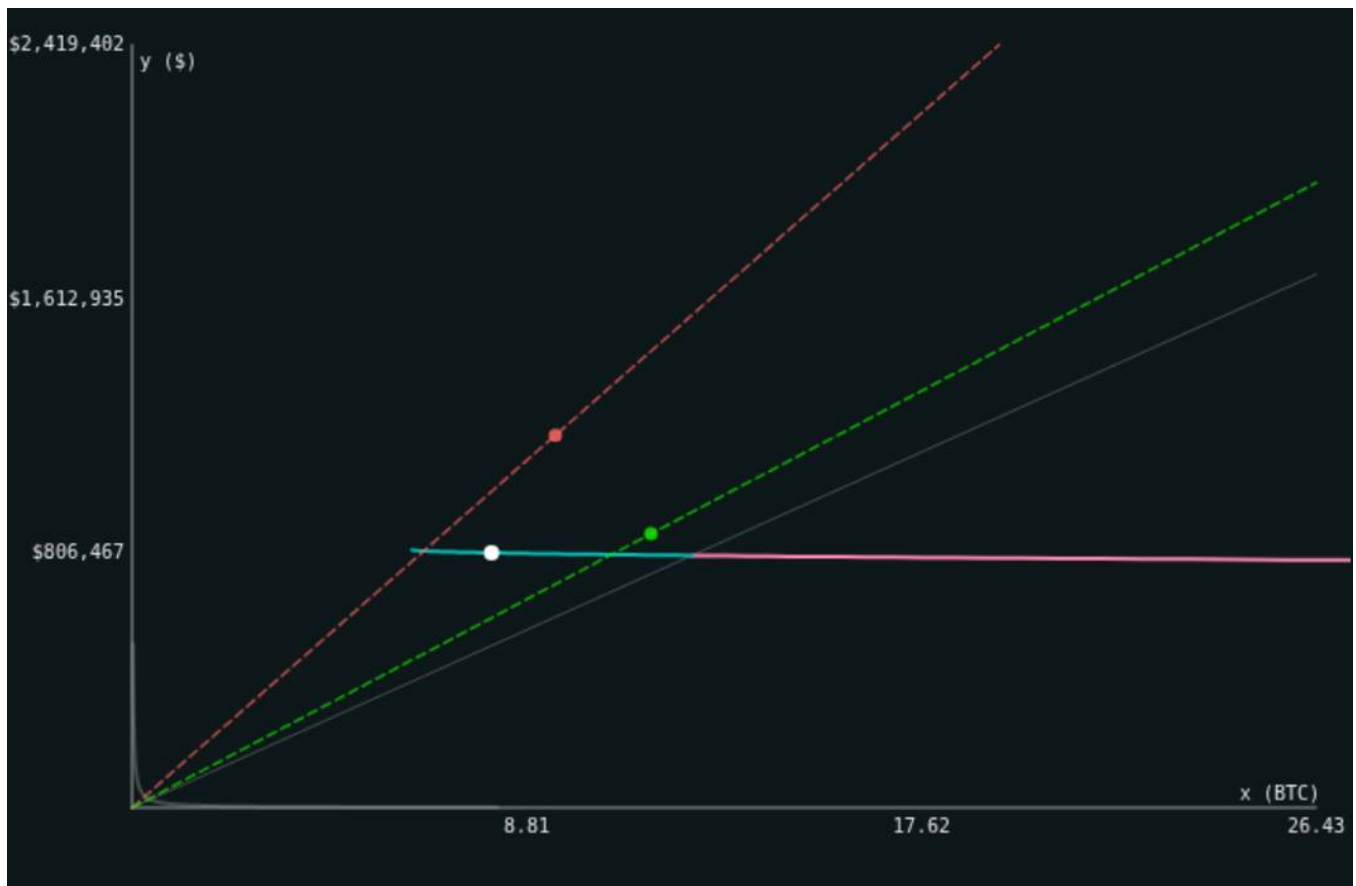
Direction Dot stays ON the curve through the trade.

Magnitude 0.00e+0 px.

Verdict: **PASS** — Trade tracking.

Step 3 — After rebase oracle \$80k → \$100k — frame may visibly shift (intended); dot stays on curve

Action: setOracleUI(100000); the frame re-fits to the new equilibrium (per slipfix change). Re-measure dot-to-curve + eq direct check.



Post-rebase: dot-to-curve $0.000e+0$ px; eq direct re-arb on-curve = true. Frame xMax before rebase = 30.00, after rebase = 26.43 (re-fit as intended by slifix; NOT a regression).

Direction Dot AND eq marker both stay on the curve through the rebase. Frame re-fit visible but the invariant (on-curve) holds.

Magnitude dot $0.00e+0$ px; eq direct true; frame xMax 30.00→26.43.

Verdict: PASS — Rebase + frame re-fit.

WORKFLOW VERDICT: PASS

Curve invariants through baseline + trade + rebase: dot on-curve ($6.42e-12$ / $0.00e+0$ / $0.00e+0$); eq on-curve via direct re-arb (true baseline, true post-rebase). Frame re-fit observed (xMax 30.00→26.43) — intended slifix behaviour, not a regression. All curve checks PASS.

D — Health: no console errors + trades execute & settle cleanly

Workflow verdict: **PASS**

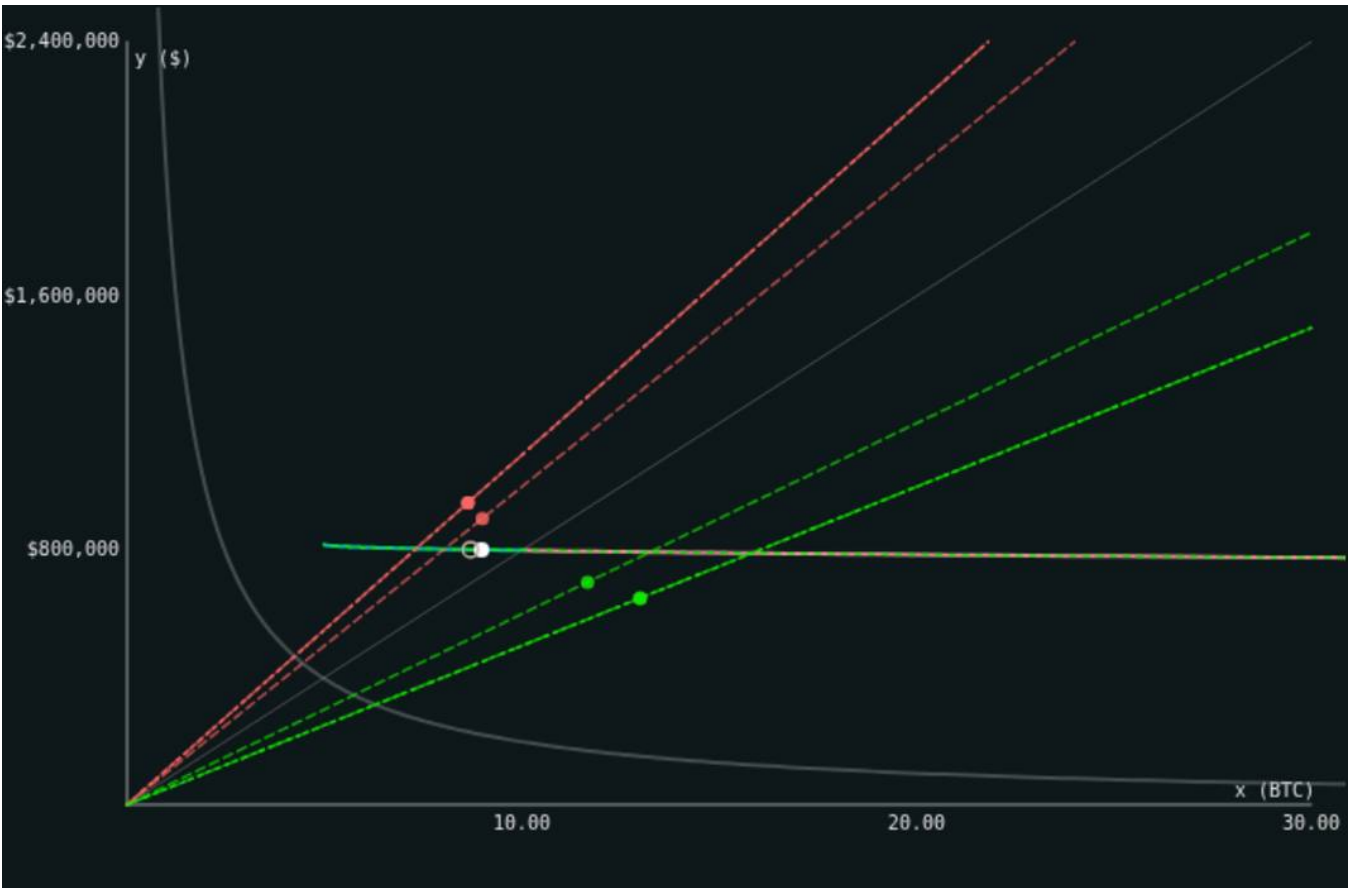
Brief Check D: no red console errors across the session; a few trades execute and settle (no frozen UI, no thrown exceptions).

Setup. Bands subtab: execute 2 band trades (sizes 0.01 and 0.005 with different strikes), sweep Perps/Settings/Portfolio subtabs, end on Transact > Bands.

Flag (read first): Errors: after nav=0, after trade 1=0, after trade 2=0, after subtab sweep=0. Pool changes: clean→{"x":9.275084288183137,"y":801353.573005574}→{"x":8.982867919750868,"y":801931.4884819709}. Portfolio band rows after 2 trades: 2.

Step 1 — Execute 2 trades + full subtab sweep + Portfolio

Action: Trade 1: sold-call \$100k/bought-put \$60k 0.01 BTC, execute. Trade 2: sold-call \$110k/bought-put \$50k 0.005 BTC, execute. Sweep Perps/Settings/Portfolio. Count band rows.



After 2 trades and full sweep: total console errors = 0. Portfolio band rows = 2. Pool reserves moved from clean (10, 800000) → (9.2751, 801353.57) → (8.9829, 801931.49).

Direction No console errors, trades settled, UI responsive.

Magnitude errors=0, settled=true, bandRows=2.

Verdict: **PASS** — Clean run.

WORKFLOW VERDICT: PASS

Console errors: 0. Trades settled: true. Portfolio shows 2 band rows after 2 trades. PASS.